Week 1+2

1

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Exercise 1.0.1

A nonassociative algebra $(A, \circ)$ is called ***left symmetric*** if the associators are left symmetric in the sense that

$$(x, y, z) = (y, x, z) \quad ((x, y, z) := (x \circ y) \circ z - x \circ (y \circ z))$$

for all $x, y, z \in A$. Prove that for every left symmetric algebra $(A, \circ)$, $(A, [\cdot, \cdot])$ is a Lie algebra, where

$$[\cdot, \cdot] : A \times A \to A, \quad (x, y) \mapsto [x, y] := x \circ y - y \circ x$$

denotes the commutator on A.

Proof. To show $[\cdot, \cdot]$ is bilinear, we see

$$\begin{aligned} [x + y, z] &= (x + y) \circ z - z \circ (x + y) \\ &= (x \circ z - z \circ x) + (y \circ z - z \circ y) \\ &= [x, y] + [y, z] \end{aligned}$$

Similarly

$$\begin{aligned} [x, y + z] &= x \circ (y + z) - (y + z) \circ x \\ &= (x \circ y - y \circ x) + (x \circ z - z \circ x) \\ &= [x, y] + [y, z] \end{aligned}$$

Futhermore, it is obvious that

$$[x, x] = x \circ x - x \circ x = 0$$

Then we only need to show the Jacobi identity. Since $(A, \circ)$ is left symmetric, we have

$$
\begin{aligned}
&[x\,[yz]] + [y\,[zx]] + [z\,[xy]] \\
&= x\circ(y\circ z - z\circ y) - (y\circ z - z\circ y)\circ x \\
&+ y\circ(z\circ x - x\circ z) - (z\circ x - x\circ z)\circ y \\
&+ z\circ(x\circ y - y\circ x) - (x\circ y - y\circ x)\circ z \\
&= x\circ(y\circ z) - x\circ(z\circ y) - (y\circ z)\circ x + (z\circ y)\circ x \\
&+ y\circ(z\circ x) - y\circ(x\circ z) - (z\circ x)\circ y + (x\circ z)\circ y \\
&+ z\circ(x\circ y) - z\circ(y\circ x) - (x\circ y)\circ z + (y\circ x)\circ z \\
&= (x\circ(y\circ z) - (x\circ y)\circ z) - (x\circ(z\circ y) + (x\circ z)\circ y) \\
&- ((y\circ z)\circ x + y\circ(z\circ x)) + ((z\circ y)\circ x - z\circ(y\circ x)) \\
&- (y\circ(x\circ z) - (y\circ x)\circ z) - ((z\circ x)\circ y - z\circ(x\circ y)) \\
&= 0
\end{aligned}
$$

Thus $(A, [\cdot,\cdot])$ is a Lie algebra. □

Exercise 1.0.2

Write $\mathrm{Der}(A)$ for the Lie algebra of derivations on A, where $A = \mathbb{C}[t, t^{-1}]$. Prove that $\mathrm{Der}(A)$ has a basis $\{L_n := t^{n+1}\frac{d}{dt} \mid n \in \mathbb{Z}\}$ and the Lie brackets among the basis elements are as follows:

$$[L_n, L_m] = (m-n)L_{m+n} \quad (m, n \in \mathbb{Z}).$$

Proof. Let $D \in \mathrm{Der}\,(A)$ be any derivation on A. Then

$$
\begin{aligned}
D\,(1) &= D\left(t\cdot t^{-1}\right) \\
&= D\,(t)\cdot t^{-1} + tD\left(t^{-1}\right)
\end{aligned}
$$

The other hand

$$D\,(1) = D\,(1\cdot 1) = D\,(1)\cdot 1 + 1\cdot D\,(1) = 2D\,(1) \implies D\,(1) = 0$$

Thus we have

$$D\,(t)\cdot t^{-1} + tD\left(t^{-1}\right) = 0 \implies D\left(t^{-1}\right) = -t^{-2}D\,(t)$$

From the linearity and the product rule, we observe that a dirivation $D \in \mathrm{Der}\,(A)$ is totally determined by its value on t, that is for any $D_1, D_2 \in \mathrm{Der}\,(A)$ if $D_1\,(t) = D_2\,(t)$, then $D_1 = D_2$. It is straightforward to verify that L_n is a derivation on A for each $n \in \mathbb{Z}$. Note that

$$L_n(t) = t^{n+1}, \quad n \in \mathbb{Z}.$$

Since $D(t) \in A$, we have

$$D(t) \in \operatorname{span}(\cdots, L_{-2}(t), L_{-1}(t), L_0(t), L_1(t), L_2(t), \cdots).$$

Then we get

$$D \in \operatorname{span}(\cdots, L_{-2}, L_{-1}, L_0, L_1, L_2, \cdots).$$

Suppose we have a finite linear combination of the L_n that equals the zero derivation: $\sum_{n\in I} k_n L_n = 0$, where $I \subset \mathbb{Z}$ is a finite set and $k_n \in \mathbb{C}$. To check whether this is the zero derivation, we can test its action on the generator t. $\left(\sum_{n\in I} k_n L_n\right)(t) = 0(t) = 0$. This gives: $\sum_{n\in I} k_n L_n(t) = \sum_{n\in I} k_n t^{n+1} = 0$. The set of monomials $\{t^j\}_{j\in\mathbb{Z}}$ is a basis for A as a $\mathbb{C}$-vector space, so they are linearly independent. The equation $\sum_{n\in I} k_n t^{n+1} = 0$ implies that all coefficients must be zero, i.e., $k_n = 0$ for all $n \in I$. This proves that the set $\{L_n\}_{n\in\mathbb{Z}}$ is linearly independent. Combining above all , we know $\operatorname{Der}(A)$ has a basis $\{L_n : n \in \mathbb{Z}\}$.

To show $[L_n, L_m] = (m-n) L_{m+n}$, we only need to show that they agree on t:

$$\begin{aligned}
[L_n, L_m](t) &= (L_n(L_m(t)) - L_m(L_n(t))) \\
&= \left(L_n\left(t^{m+1}\right) - L_m\left(t^{n+1}\right)\right) \\
&= \left((m+1)\, t^{m+n+1} - (n+1)\, t^{m+n+1}\right) \\
&= (m-n)\, t^{m+n+1} \\
&= (m-n)\, L_{m+n}(t)
\end{aligned}$$

which completes the proof.

□

Exercise 1.0.3

Let $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and let V be an ***n*-dimensional** vector space over $\mathbb{F}$ equipped with a nondegenerate bilinear form

$$B : V \times V \to \mathbb{F}, \quad (v, w) \mapsto B(v, w).$$

Prove that:

(a) If $\mathbb{F} = \mathbb{C}$ and B is symmetric, then there exists a basis $\{v_1, v_2, \ldots, v_n\}$

of V such that $B(v_i, v_j) = \delta_{i,j}$ for all $1 \le i, j \le n$.

(b) If $\mathbb{F} = \mathbb{R}$ and B is symmetric, then there exists a basis $\{v_1, v_2, \dots, v_n\}$ of V and $p = 0, 1, \dots, n$ such that $B(v_i, v_j) = \varepsilon_i \delta_{i,j}$ for all $1 \le i, j \le n$, where $\varepsilon_i = 1$ if $i \le p$ and $\varepsilon_i = -1$ if $i > p$.

(c) If $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$ and B is skew-symmetric, then $n = 2m$ is even and there exists a basis $\{v_1, v_2, \dots, v_n\}$ of V such that the matrix

$$(B(v_i, v_j))_{1 \le i,j \le n} = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix},$$

where I_m denotes the identity matrix of rank m.

Proof. Lemma: If V is a vector space on the field $\mathbb{K}$ satisfying $\mathrm{char}\,(K) \neq 0$, B is a nonzero symmetric bilinear form on V , then there exists $v \in V$ such that $B\,(v, v) \neq 0$.

Proof of Lemma: Otherwise, we have for each $w, v \in V$,

$$\begin{aligned} 0 &= B\,(w + v, w + v) \\ &= B\,(w, w) + B\,(w, v) + B\,(v, w) + B\,(v, v) \\ &= 2B\,(w, v) \end{aligned}$$

which leads to a contradiction since B is nondegenerate.

(a) We induct on the dimension n. If $n = 1$, there exists vector v_0 such that $B\,(v_0, v_0) \neq 0$. We denote $B\,(v_0, v_0)$ by $\lambda \in \mathbb{C}$. λ has a square root in C, denoted by $\sqrt{\lambda}$. Let $v_1 = \frac{v_0}{\sqrt{\lambda}}$. It constitute a basis of V such that

$$B\,(v_1, v_1) = \frac{1}{\lambda} B\,(v_0, v_0) = 1$$

Now we suppose that $(n-1)$-dimensional space has property (a). We consider the n-dimensional vector space V on $\mathbb{C}$, with a nondegenerate bilinear form B. By the Lemma, we can take a vector v_0 such that $B\,(v_0, v_0) \neq 0$, and $v_1 := \frac{v_0}{\sqrt{B(v_0, v_0)}}$ such that $B\,(v_1, v_1) = 1$. Now we define

$$W := \{w : B\,(v_1, w) = 0\},$$

which is a subspace of V. We claim that $V = W \oplus \langle v_1 \rangle$, and thus $\dim W = n - 1$.

1. $W \cap \langle v_1 \rangle = 0$: Take any $w \in W \cap \langle v_1 \rangle$, then there is a complex valued c such that $w = cv_1$. Futhermore,

   $$0 = B(w, v_1) = cB(v_1, v_1) = c \implies c = 0$$

   There must be $w = 0$.

2. $W + \langle v_1 \rangle = V$: Take any $v \in V$. Let $w = v - B(v, v_1) v_1$, then

   $$B(w, v_1) = B(v - B(v, v_1) v_1, v_1) = B(v, v_1) - B(v, v_1) = 0$$

   Thus $w \in W$, $v = w + B(v, v_1) v_1$. Therefore $v \in W + \langle v_1 \rangle$.

Now we prove that $B|_W : W \times W \to \mathbb{C}$, $B|_W(w_1, w_2) \coloneqq B(w_1, w_2)$ is a nondegenerate bilinear form on W.

1. Nondegenerate: For $w_0 \in W$, if for any $w \in W$, there is $B|_W(w_0, w) = 0$, then for any $v \in V$, set $v = k_1 v_1 + k_2 w_1$, $k_1, k_2 \in \mathbb{C}, w_1 \in W$. We have

   $$B(w_0, v) = k_1 B(w_0, v_1) + k_2 B(w_0, w_1) = 0 + 0 = 0$$

   Since B is nondegenerate, $w_0 = 0$. Therefore $B|_W$ is nondegenerate.

2. Symmetric and bilinear: Since B is symmetrc and bilinear, it is obvious that $B|_W$ is as well.

From our inductive assumption, ther exists a basis $\{v_2, \cdots, v_n\}$ for W such that $B(v_i, v_j) = \delta_{i,j}, 2 \le i, j \le n$. Then $\{v_1, \cdots, v_n\}$ is a basis for V such that $B(v_i, v_j) = \delta_{i,j}, 1 \le i, j \le n$.

(b) We induct on the dimension n. As in the (a), if $n = 1$, there exists vector v_0 such that $B(v_0, v_0) \neq 0$. We still denote $B(v_0, v_0)$ by λ , but here $\lambda \in \mathbb{R}$. We set

$$v_1 = \frac{v_0}{\sqrt{|\lambda|}}$$

Then

$$B(v_1, v_1) = B\left(\frac{v_0}{\sqrt{|\lambda|}}, \frac{v_0}{\sqrt{|\lambda|}}\right) = \frac{\lambda}{|\lambda|} = \begin{cases} 1, & \lambda > 0 \\ -1, & \lambda < 0 \end{cases}$$

Now we suppose that any $(n-1)$-dimensional space has property (b). We consider the n-dimensional vector space V on $\mathbb{R}$. A smililar manner as above gives a vector u_1 such that $B(u_1, u_1) = \pm 1 =: \varepsilon_1$. Still , define

$$W \coloneqq \{w : B(u_1, w) = 0\}$$

After a same process as in (a), we have $V = W \oplus \langle u_1 \rangle$, $\dim W = n-1$, $B|_W$ is a nondegenerate bilinear form on W. From our inductive assumption, there exists a basis $\{u_2, \cdots, u_n\}$ for W such that $B\left(u_i, u_j\right) = \varepsilon_i \delta_{i,j}, \varepsilon_i \in \{-1, 1\}, 2 \le i, j \le n$. Then $\{u_1, \cdots, u_n\}$ is a basis for V such that $B\left(u_i, u_j\right) = \varepsilon_i \delta_{i,j}, \varepsilon_i \in \{-1, 1\}, 1 \le i, j \le n$. Finally , we can relable $\{u_1, \cdots, u_n\}$ to be $\{v_1, \cdots v_n\}$ such that there exists $p \in \{0, 1, \cdots, n\}$ such that $B\left(v_i, v_j\right) = \varepsilon_i \delta_{i,j}$ for all $1 \le i, j \le n$, where $\varepsilon_i = 1$ if $i \le p$ and $\varepsilon_i = -1$ if $i > p$.

(c) For each $v \in V$, $B(v, v) = -B(v, v) \implies B(v, v) = 0$ since $\operatorname{char}(\mathbb{F}) \neq 2$. Let V be vector space with arbitray positive dimension on $\mathbb{F}$. Claim that exists vector $v_0, w_0 \in V$ such that $B(v_0, w_0) \neq 0$. Otherwise, for each $w, v \in V$,

$$B(w, v) = B(v, w) = 0$$

, which is a contradiction to B is nondegenerate. We set

$$v_1 = v_0, \quad w_1 = \frac{w_0}{B(v_0, w_0)}$$

Then $B(v_1, w_1) = 1$. Define

$$W \coloneqq \{w : B(w, \operatorname{span}(v_1, w_1)) = 0\}$$

Then

1. $W \cap \operatorname{span}(v_1, w_1) = 0$: Take any $u \in W \cap \langle v_1, w_1 \rangle$, then there exists $m, n \in \mathbb{F}$ such that $u = mv_1 + nw_1$. Then

   $$B(u, v_1) = B(mv_1 + nw_1, v_1) = nB(w_1, v_1) = 0 \implies n = 0$$

   $$B(u, w_1) = B(mv_1 + nw_1, w_1) = mB(v_1, w_1) = 0 \implies m = 0$$

   Thus $u = 0$.

2. $W + \operatorname{span}(v_1, w_1) = V$: Take any $v \in V$, let $w = v - B(v, w_1)v_1 - B(v_1, v)w_1$. Then

   $$B(w, v_1) = B(v - B(v, w_1)v_1 - B(v_1, v)w_1, v_1) = B(v, v_1) + B(v_1, v) = 0$$

   $$B(w, w_1) = B(v - B(v, w_1)v_1 - B(v_1, v)w_1, w_1) = B(v, w_1) - B(v, w_1) = 0$$

   Thus we have $V = W \oplus \operatorname{span}(v_1, w_1)$. $\dim W = \dim V - 2$.

   Now we claim that the dimension of V must be even. Otherwise, once we get $V = W \oplus \operatorname{span}(v_1, w_1)$, we can do the same thing for W and $B|_W$, and get $W = W_1 \oplus \operatorname{span}(v_2, w_2)$. But after finitely many decomposition, we get

$$V = \operatorname{span}(v_1, w_1) \oplus \cdots \oplus \operatorname{span}(v_m, w_m) \oplus W_{m-1}$$

such that $\dim W_{m-1} = 1$. Now there is no vector v_m, w_m such that $B|_{W_{m-1}}(v_m, w_m) = B(v_m, w_m) > 0$, which is a contradiction. Thus there exists $m \in \mathbb{Z}_{>0}$ such that $\dim V = 2m$. We have shown above that there exists $v_1, \cdots, v_m, w_1, \cdots, w_m \in V$ such that

$$V = \operatorname{span}(v_1, w_1) \oplus \cdots \oplus \operatorname{span}(v_m, w_m)$$

And $B(v_i, w_i) = 1, 1 \leq i \leq m$. $B\left(v_i, w_j\right) = 0, i \neq j$. We relabel

$$\{v_1, \cdots, v_m, w_1, \cdots, w_m\}$$

to be

$$\{v_1, \cdots, v_m, v_{m+1}, \cdots, v_n\}$$

Then we have

$$B(v_i, v_{i+m}) = 1, \quad B(v_{i+m}, v_i) = -1, \quad i = 1, 2, \cdots, m$$

and

$$B\left(v_i, v_{j+m}\right) = 0, \quad 1 \leq i \neq j \leq m$$

Thus we have the matrix

$$(B(v_i, v_j))_{1 \leq i,j \leq n} = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}.$$

$\square$

Index

Exercises